

Decoding AI Diffusion

Mapping the path
of transformative AI
across industries

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PUBLISHED

JULY 2025



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Abstract

In this report, we develop an organizing framework to assess the diffusion potential of transformative AI (TAI) across different sectors of the economy. We argue that many existing methodologies for assessing TAI's potential to augment and automate cognitive labor oversimplify the structural, cultural, and technological differences between industries which will impact their susceptibility to disruption from AI. Drawing upon TAI literature and historical diffusion patterns of past digital general purpose technologies, we identify the sectoral factors most likely to impact the breadth and speed of TAI diffusion. We propose a five-category framework to organize these factors: 1) technological readiness, 2) workforce and human capital, 3) investment, 4) markets and competition, 5) regulation and oversight. This structure can be applied sectorally, leveraging qualitative and quantitative indicators for a given sector in order to determine the likely path and speed of TAI within that sector, or to compare TAI's potential impact across different sectors. We aim to provide policymakers, business leaders, and researchers with a tool to assist more nuanced foresight, enabling stakeholders to better anticipate labor market shifts from AI.

Executive Summary

1. The Challenge: Understanding Transformative AI's Uneven Economic Integration

Transformative AI (TAI)—artificial intelligence systems with the potential to cause societal shifts comparable to past industrial revolutions—is rapidly advancing. While discourse often focuses on when TAI might arrive or which tasks it could automate, there's a critical underestimation of the complexities surrounding its economic integration. Prevailing assumptions of rapid, uniform adoption overlook significant historical precedents and inherent sectoral differences. This paper argues that TAI diffusion will be an uneven, relatively gradual process, with adoption patterns varying significantly across economic sectors, organizations, and job roles, particularly within knowledge work.

The critical challenge lies in moving beyond generalized predictions to a more nuanced understanding of these diffusion pathways. Such understanding is paramount for policymakers, business leaders, and researchers to anticipate labor market shifts, design targeted interventions, and develop effective AI governance. To address this, our research introduces a structured, five-category framework designed to assess the varying susceptibility of different economic sectors to TAI diffusion, based on their unique structural, technological, economic, and cultural characteristics.

2. The Core Contribution: A Five-Category Framework for Sectoral TAI Diffusion Analysis

Our research proposes a qualitative framework that organizes the key determinants of industry-level AI diffusion into five overarching categories. This framework moves beyond task-level automation predictions to offer a more holistic, sectoral analysis, enabling a systematic evaluation of why certain industries may lead in TAI adoption while others lag.

- **I. Technological Readiness:** Assesses an industry's existing digital foundation and capabilities to implement AI.
 - **Digital Maturity (+):** Higher existing technological sophistication (data, IT systems, digital processes) lowers adoption barriers.
 - **AI Ecosystem Density (+):** Greater availability of relevant AI tools, models, and frameworks accelerates diffusion.
 - **Complementary Technology Required (-):** High dependence on other technologies (e.g., IoT, robotics, advanced networks) can create bottlenecks if those technologies are immature or not widely adopted within the sector.
- **II. Workforce and Human Capital:** Examines the social, cultural, and

human skill elements influencing AI adoption.

- **Cultural Resistance to Change (-):** Entrenched traditional norms, risk aversion, and hierarchical structures slow diffusion.
- **AI Skills (+):** A strong talent pool (data scientists, ML engineers, AI-literate managers) increases diffusion rates; shortages are a bottleneck.
- **Labor Power (-):** Strong labor representation can slow or alter diffusion through negotiations over job security, retraining, and implementation terms.
- **III. Investment:** Considers the financial dynamics and capital flows driving AI adoption.
 - **Private AI Investment (+):** Significant venture capital, corporate R&D, and private equity funding for AI accelerate innovation and diffusion.
 - **Public AI Investment (+):** Government funding, subsidies, and incentives can de-risk research and promote adoption, especially in strategic sectors.
 - **AI Capital Expenditures (+):** Direct industry investment in foundational AI capabilities (e.g., specialized compute infrastructure, AI talent acquisition) directly fuels diffusion.
- **IV. Markets and Competition:** Analyzes the competitive landscape, labor market conditions, and demand dynamics.
 - **Market Competitiveness (+):** Intense rivalry compels firms to adopt AI for efficiency gains or competitive advantage.
 - **Labor Market Tightness (+):** Worker shortages and rising wage costs make AI automation or augmentation more economically attractive.
 - **Demand (+):** Strong and growing customer demand for an industry's products/services, especially for AI-enabled features, is a powerful accelerator for AI investment and diffusion.
- **V. Regulation and Oversight:** Evaluates the impact of the governance structures and legal frameworks.
 - **Regulatory Burden (-):** Complex, stringent, or costly existing non-AI-specific regulations can slow AI diffusion by increasing compliance overhead.
 - **Enforcement and Oversight Strength (-):** While predictable enforcement fosters trust, increasing scrutiny, severe penalties, or unclear oversight can heighten perceived risks and slow adoption.
 - **AI-Specific Regulations (-):** New laws explicitly targeting AI (e.g., data governance, algorithmic accountability) typically slow initial diffusion as firms adapt and invest in compliance, though they can also provide clarity in the long run.

This framework provides a memorable and structured approach for stakeholders to analyze the multifaceted forces governing TAI diffusion,

facilitating a more nuanced understanding of industry-specific adoption trajectories.

3. Framework in Action: Contrasting High-Frequency Trading and Retail Banking

The utility of the framework is demonstrated by comparing TAI diffusion prospects in High-Frequency Trading (HFT) and Retail Banking:

- **High-Frequency Trading (HFT): Exceptional Susceptibility and Rapid Transformation** HFT is poised for rapid and comprehensive TAI diffusion.
 - **Key Enablers (reflecting framework categories):** Exceptional *Digital Maturity* and established *AI Capital Expenditures* (Technological Readiness, Investment Factors); extreme *Market Competitiveness* driving innovation (Market and Competitive Factors); and high availability of specialized *AI Skills* (Workforce and Human Capital).
 - **Primary Bottleneck:** Potential Regulatory Burden related to the explainability and oversight of AI-driven trading algorithms (Regulation and Oversight). TAI is expected to automate most labor, focusing human roles on high-level strategy and risk oversight.
- **Retail Banking: Moderate Susceptibility and Uneven, Protracted Transformation** Retail banking faces a more moderate, uneven, and longer TAI diffusion timeline.
 - **Key Enablers:** Strong *Demand* for personalized digital experiences and ongoing *AI Capital Expenditures* (Market and Competitive Factors, Investment Factors).
 - **Primary Bottlenecks:** Significant *Legacy IT Systems* hampering *Digital Maturity* (Technological Readiness); high *Regulatory Burden* (Regulation and Oversight); *Cultural Resistance to Change* in traditional institutions; and relatively lower *AI Skills* density compared to other financial sectors (Workforce and Human Capital). TAI is likely to augment many customer-facing and advisory roles while automating back-office tasks, with a slower transformation of core banking systems.

Comparative Insights: The stark contrast highlights that TAI diffusion speed and impact are not solely determined by the technology's potential but are fundamentally shaped by an industry's unique configuration across the five framework categories. HFT's digital-native structure and intense competitive dynamics create an ideal environment for rapid adoption, whereas retail banking's legacy systems, regulatory complexity, and cultural inertia lead to a more measured and phased approach.

4. Implications and Strategic Imperatives for a TAI-Driven Future

Understanding the uneven and sector-specific nature of TAI diffusion is critical for effective preparation. This framework offers several key implications:

- **For Policymakers:** Enables the design of targeted interventions, such as sector-specific skill development programs, infrastructure investments, or adaptive regulatory sandboxes, rather than one-size-fits-all approaches. It also informs where regulatory attention might be most urgently needed to manage risks or foster innovation.
- **For Business Leaders:** Provides a tool for strategic foresight, helping to identify sector-specific opportunities and threats, guide investment decisions in AI and complementary capabilities, and manage workforce transitions proactively.
- **For Researchers:** Highlights the need for more granular, sector-level data collection on AI adoption and its impacts. It also opens avenues for future quantitative analysis, such as developing weighted indices for the framework factors to enhance predictive power.

The diffusion of TAI will not be an overnight revolution but an evolving process of integration, disruption, and adaptation that varies significantly by industry.

We will likely only recognize the full transformative impact of TAI in retrospect. Therefore, a proactive and nuanced understanding of its likely diffusion pathways, as facilitated by this sectoral framework, is essential. By better anticipating how TAI will reshape the economic order, stakeholders can make more informed decisions, mitigate negative consequences, and harness the benefits of this powerful technology for broader economic well-being. Advancing data collection efforts and fostering a grounded, evidence-based dialogue on AI diffusion are paramount as we navigate this period of profound technological change.

Introduction

Transformative AI: Impacts and Adoption

The rapid pace of AI development has caused researchers, firms, economists, and policy makers to consider the real possibility that transformative AI (TAI) systems will soon be developed. TAI acknowledges AI's potential to reshape society even before reaching artificial general intelligence (AGI) or human-level AI (HLAI), providing a useful framing for policymakers, researchers, and organizational leaders preparing for the concrete ways that AI might reshape labor markets, governance systems, and social institutions in the short to medium term. Experts' predictions of the timeline to achieve these systems vary, though many agree that these systems may be developed within the next ten years, if not sooner^{1,2,3}. Many leading technologists, including those developing frontier generative AI models, predict that TAI will be reached within 2-3 years^{4,5}, though it is important to maintain some skepticism towards their predictions given their incentives to promote AI's progress.

In addition to research predicting when TAI systems may arise and which human tasks they will ultimately be able to accomplish, there is a nascent field of research into the economic and societal effects of TAI. Leading economists have begun researching these potential impacts, employing traditional economic frameworks to predict macroeconomic impacts from TAI. This research frequently warns of negative labor market outcomes but predicts more modest productivity increases than many predictions from AI technologists^{6,7,8,9}, demonstrating the divergence in expectations for TAI stemming from differences in their disciplines: technologists often emphasize the theoretical capabilities of TAI and rapid innovation cycles, while economists foreground institutional frictions, diffusion lags, and historical models of technology-fueled growth, which show that even highly transformative general-purpose technologies (GPTs) are not adopted overnight^{10,11}.

However, much of this research lacks deep discussion of how historical technology diffusion patterns may shape the spread of TAI or oversimplifies the complexity of potential diffusion pathways of TAI, limiting the predictive power and utility of these analyses. Our research seeks to nuance the conversation around TAI diffusion through analysis of current TAI research and historical technology diffusion patterns. We assert that TAI's diffusion, which will occur unevenly across regions, industries, organizations, and job roles, is an under-discussed component of the literature on TAI which must be developed further in order to better assess the likely economic and societal consequences of its development and deployment. By drawing on historical analogues and contemporary data, our work aims to highlight how contextual

factors will mediate the real-world impacts of TAI, ultimately shaping who will benefit and who is displaced as the technology spreads.

Research Direction and Scope

Our analysis focuses on diffusion of AI systems across and throughout organizations rather than personal use of AI. We believe that while individuals' rapidly increasing adoption of AI may impact personal productivity¹², diffusion of AI across organizations will ultimately impact hiring and firing decisions which, in the aggregate, will reshape the labor economy^{13,14}.

It is unlikely that the rate of AI adoption will be equivalent across job categories at different firms, particularly as we compare roles based on the level of digitalization they exhibit and the potential applicability of AI to these roles. It is similarly unlikely that job roles and firms within different sectors, even if they exhibit otherwise similar characteristics, will experience similar levels of AI diffusion. Diffusion will also likely vary considerably across different national and regional contexts, but our research analyzes sectors through a single geographic lens.

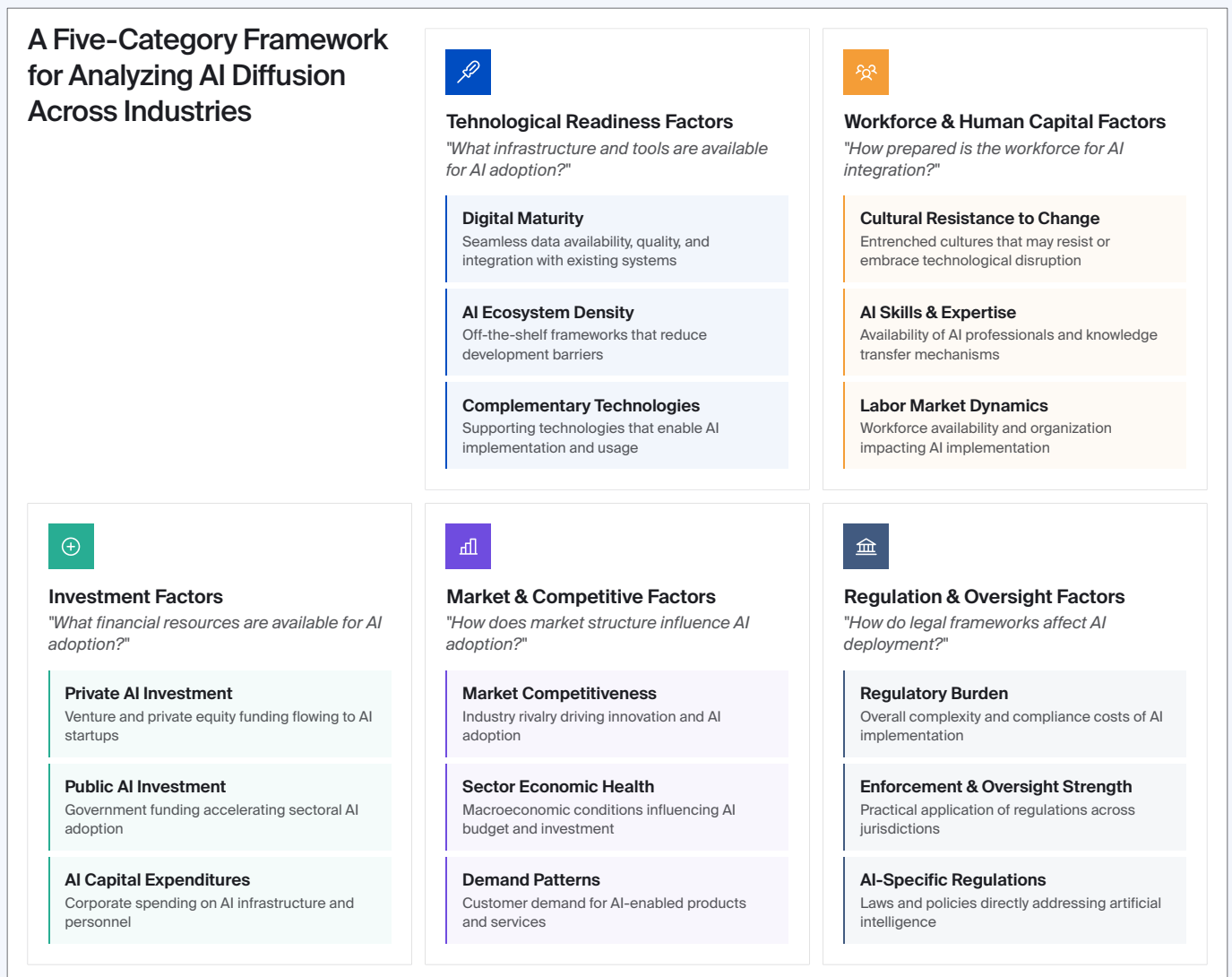
Our analysis centers around knowledge work, which today heavily leverages digital technologies both for individual tasks (data analysis, coding, research, etc.) or working in teams (video conferencing, synchronous collaboration tools, etc.). It is therefore no surprise that many current LLMs and other generative AI tools, as well as future TAI use cases, target the knowledge work sector. While future robotics and other embodied technologies may someday transform roles which center around physical tasks, given the current level of development of these technologies, we believe that it is more pressing to consider the possible diffusion paths of TAI across knowledge work roles. Within the field, we expect an uneven adoption of AI tools depending on sector and role characteristics, consistent with what we have seen to date: some knowledge work jobs have already been impacted by current AI systems^{15,16}, while others are likely to resist displacement for an extended period, with certain roles remaining exclusively human.

Sectoral Diffusion of TAI

Our research attempts to provide a framework to analyze where TAI is most likely to gain traction once developed and aggregate data sources which can be observed to track the diffusion of AI systems over time across sectors of the economy, starting today. While it is impossible to conclusively predict how TAI may impact specific tasks or job roles at individual companies, there are economic, labor, technological and regulatory factors which we believe will determine the susceptibility of different sectors of the economy to TAI diffusion. This research develops a consolidated framework of organizational categories of factors which can serve as a memorable structure for evaluating TAI's diffusion across industries based on their susceptibility to this

technology.

To build these factors, we analyzed current literature on TAI and its potential impacts, particularly on the labor economy. By synthesizing this research, we built our own model of factors from the bottom up, some outlined in the literature and some novel, which would impact diffusion of AI systems throughout organizations within and across different industry sectors. For each factor, we identified and aggregated data sources into observable indicators, allowing us to assess different sectors over time against our metrics. We then consolidated these factors into five categories which concisely outline the characteristics that will make certain sectors more or less susceptible to TAI diffusion:



Research Goals

Our research is intended to further the nascent field of research on the economic effects of TAI by developing a framework of factors to assist in measuring the susceptibility of different sectors of the US economy to automation and augmentation by AI. **We hope that this framework can be used to better predict potential rates and paths of AI diffusion across firms**

within each sector of the economy. Our framework deliberately avoids deep analysis of task-level features which may make specific job roles susceptible to displacement or augmentation by AI, and likewise does not encompass every macro-level metric which will impact AI diffusion. We hope that our focus on economic sectors will propel further research into the economic and social effects of TAI past the generalizations present in much of today's research on the economic impacts of AI.

We implore other AI researchers, firms, policy makers, and individuals to recognize and address the critical role of AI diffusion as they prepare for a TAI future. TAI is defined based on its impact on society, making it fundamentally a lagging identifier; we will not know that TAI has been achieved until it is already transforming the world around us. Because of this, **we view achieving TAI not as a single point in time, but as a period of ferment across the economy** (and society more broadly) as current jobs are augmented or replaced and new jobs are created.

By better understanding the way that organizations have adopted technologies in the past and, crucially, how they are adopting AI today, we can build a better understanding of how TAI will reshape the economic order and how fast job augmentation or displacement will occur. This understanding can help policy makers and firms to better understand potential labor market disruptions, target educational initiatives for workers and students, analyze shifting competitive dynamics across markets, and build better AI governance policies. Ultimately, the pace and pattern of TAI diffusion will impact the effect of this technology on humans' economic well-being, making this research a priority to better anticipate and navigate TAI's economic and societal impacts.

Background

Transformative AI (TAI)

Artificial intelligence has, in recent years, evolved from a research field into a mainstream technology that has attracted massive investment, proliferated across individual and business applications, and commanded increasing attention from policymakers, organizations, and the general public. Many AI researchers and science and technology professionals predict that this technology will continue improving as additional computing power and data become available and models continue to improve^{17,18,19,20}. Multiple terms are used in the AI field to describe these more powerful systems, including “human-level AI,”²¹ and artificial general intelligence (AGI), which are generally defined similarly any system that “can perform intellectual tasks as well as, or better than, a human” on some set of tasks^{22,23}. But, these terms generally refer to these systems’ capabilities rather than their impacts. For this reason, many researchers have begun to use the term “Transformative AI” (TAI) instead.

Definitions of Transformative AI (TAI) vary, but generally refer to AI systems that may generate significant societal changes^{24,25}. TAI is defined by the scale of its potential impact, with comparisons made to transformative events in human history such as the industrial or agricultural revolutions^{26,27,28}, or systems which could “lead to radical changes in welfare, wealth, or power”²⁹. This focus on impact distinguishes TAI from other definitions of powerful AI such as AGI and HLAI, which focus more on technical capabilities than the real-world impacts of those technologies. This framing helps bridge the gap between the AI safety research community and the broader scholarly community, allowing a diverse audience to meaningfully engage in discussions about AI’s future without requiring deep, technical domain expertise.

This focus on outcomes rather than capabilities also makes TAI a more useful term when analyzing the economic, social, and political effects of AI as it proliferates, specifically by highlighting that AI systems “may have profound and long-lasting impacts on society well before anything close to human-level intelligence is reached”²². This allows researchers to examine how AI capabilities might transform (or already is transforming) particular sectors, institutions, or social arrangements^{30,31,32,33}, though some of this research uses alternate terminology. This outcome-focused approach facilitates more nuanced discussions about the differential impacts of AI advances across contexts and the varied timelines on which transformations might occur.

TAI acknowledges that AI’s transformative potential exists across a spectrum, even in the absence of human-level general intelligence. This is particularly relevant for short- to medium-term analysis, as it focuses attention on the more

immediate and potentially disruptive impacts of AI that may not require full AGI to manifest. This makes the concept of TAI particularly useful for policy makers, researchers and organizational leaders preparing for the concrete ways that AI might reshape labor markets, governance systems, and social institutions^{34,35,36}. The TAI framing encourages forward-looking, concrete adaptation strategies and policy proposals to solve concrete problems likely to arise as AI proliferates^{37,38}.

Labor Market Effects of TAI

Within the literature, there exists a wide range of beliefs regarding AI's impact on the labor market, but the balance of research suggests changes to the future of work will have major macroeconomic effects. Evolving AI systems have the potential to generate massive economic value^{39,40}, potentially increasing productivity and GDP growth rapidly^{41,42,43}. These macroeconomic effects will likely also have major ramifications for the labor market. IMF research shows “almost 40 percent of global employment is exposed to AI,” with up to 60 percent of employment in advanced economies exposed⁴⁴. A widely-cited article in 2017 claimed that in the US, 47% of jobs are exposed to AI by 2030⁴⁵.

The potential impacts of AI on the labor market impacts will vary heavily based on whether jobs will be automated, and therefore replaced by AI, or augmented, which can instead increase human productivity while preserving employment. IMF researchers argue that knowledge worker exposure from AI may, on net, have positive labor market benefits, with “college educated individuals...poised to reap AI benefits” and the potential for income to surge for all workers if productivity gains are high enough⁴⁶. A report from the Tony Blair institute argues that labor substitution and increased demand from new technologies generally tend to balance out, meaning that unemployment from AI will be offset by new roles working on AI-enabled products and services pulling workers back into the workforce⁴⁷.

Not all researchers agree, with some predicting that AI automation will outweigh augmentation and lead to higher unemployment levels overall. Gmyrek, Berg and Bescond⁴⁸ project overall unemployment to rise modestly, but argue that uneven AI diffusion will specifically impact certain sectors much more significantly. It is also critical to consider the period of transition to AI and its impact on unemployment; even if the equilibrium outcome of AI is steady or increased employment levels, jobs destruction outpacing rehiring in the short term can negatively impact workers⁴⁹.

Some have attempted to forecast the impact of AI on unemployment specifically, but this task proves more difficult because unemployment will ultimately depend on policy interventions (or lack thereof), the rate of AI adoption, and the rate of technological progress of frontier AI systems. Frank, Ahn and Moro⁵⁰ modeled potential labor market impacts of AI based on job role exposure across a variety of methodologies and scenarios, and determined

that unemployment risk can be predicted in many cases using these methods, though they acknowledged potential factors which could confound the direct relationship between AI proliferation and labor market effects. Researchers polled as part of the Convergence Threshold 2030 Summit predicted a range of unemployment scenarios within specific sectors, with some predicting relatively stable unemployment rates by 2030 but the potential for massive unemployment in the future if AI is able to accomplish entire task groupings that make up a given job or sector⁵¹.

The speed and scope of AI's diffusion will have major impacts on the level of labor market disruption⁵². If TAI takeoff is primarily constrained technologically, for example by limitations on access to computing infrastructure and models' recursive learning capabilities⁵³, then the period of transition to TAI could cause wide-ranging, rapid displacement of human employment. While we acknowledge the importance of these technological factors to TAI's development, we align with the perspective advanced by Narayanan and Kapoor⁵⁴, who argue that "the speed of diffusion is inherently limited by the speed at which not only individuals, but also organizations and institutions, can adapt to technology." This perspective foregrounds the need to consider diffusion as a key factor in developing more plausible future scenarios regarding the economic impacts of TAI.

Historical Technological Diffusion

Outside of the specific field of AI, there is a wealth of research exploring the theory of how technologies diffuse. Many have also studied and analyzed the adoption patterns of past general purpose technologies (GPTs) and digital technologies. We leverage this work to enrich the discussion of how increasingly powerful AI systems may diffuse by understanding key factors which have propelled or slowed the adoption of other technologies. We will also explore the nascent literature analyzing which factors are most heavily impacting AI's diffusion thus far, building these factors into our framework.

The literature on technological diffusion is vast and conclusions vary considerably based on the institutional context and the characteristics of the technology being analyzed. For the purposes of this research, we will focus briefly on the theories of diffusion of general purpose technologies (GPTs). GPTs are characterized by their pervasive impact on the economy; GPTs become inputs in downstream sectors' R&D processes and increase the productivity of the economy as a whole⁵⁵. These technologies also have a large capacity for improvement and their value can be extended by complementary technologies⁵⁶.

The internet stands as a quintessential example of a GPT, demonstrating the characteristics above. Since its widespread adoption in the 1990s, it has fundamentally transformed virtually every sector of the global economy. As predicted by GPT theory, the internet has become a critical input in

downstream innovation processes, enabling entirely new business models and research capabilities that have accelerated productivity growth across diverse industries. The internet's architecture has shown remarkable capacity for improvement, evolving from dial-up connections to high-speed broadband and now to mobile and satellite networks. Its value has been dramatically extended through complementary innovations such as cloud computing, mobile applications, and artificial intelligence systems that leverage its connectivity. Due to the transformative potential of AI systems, some have already classified AI, specifically generative AI, as a GPT⁵⁷.

Helpman and Trajtenberg⁵⁸ assert that widespread adoption of GPTs may occur considerably later than their invention, and that sector-specific complementary innovations generally increase productivity in those sectors rather than simply adopting the GPTs themselves. This extended period of diffusion and complementary innovation can lead to lags in measurable productivity gains from GPTs as well, which Brynjolfsson⁵⁹ dubbed the “productivity paradox.”

Narayanan and Kapoor⁶⁰ contend that AI's diffusion will diffuse in a way similar to previous GPTs over “decades, not years”. They claim that one must look beyond capability benchmarks and instead focus on measuring progress in real-world AI applications, which have thus far lagged the seemingly rapid technological progress of AI systems when measured against benchmarks like the bar exam or self-contained coding assessments. But many others disagree with this claim that economic impacts will be gradual, arguing instead that productivity gains from AI will not cause productivity to rise more rapidly than with previous GPTs^{61,62,63}.

To build a framework to better understand how AI may diffuse across sectors, we have explored previous literature to determine sectoral factors which have impacted the spread of past transformative technologies, with particular focus on digital technologies. Research has found a “large and growing gap” in digitalization—defined as leveraging digitized data and tools to transform a business around the new opportunities and demands surfaced by digital technology⁶⁴—across different sectors of the US economy^{65,66}. Gandhi, Khanna and Ramaswamy⁶⁷ categorize different factors which impact technology diffusion into three broad categories: digitized physical assets, digital usage across business processes, and digitally empowered workforces. Olmstead⁶⁸ breaks each high-level factor into measurable subfactors; digital assets consist of digital spending and digital asset stock, digital usage describes transactions, interactions, business processes, and market mining while digital labor can be broken down into digital spend on workers and tools. Analysis of these factors shows that industries with more digitally enabled workforces are disproportionately more likely to outpace other sectors in adopting new digital technologies, with leading sectors having workforces which are “13 times more digitally enabled” than laggards⁶⁹. This research was conducted almost ten years ago so it is possible some of the gaps across these digitization factors have tightened, but the fundamental insights into the drivers of increased

digitization provide us a useful basis when developing our key AI diffusion factors.

Sabbagh *et al.*⁷⁰ argue that sectoral adoption of digital technologies is impacted by the interaction between a technology and four key business functions (business models, go-to-market, production, and operations) in each sector. For example, a sector where increased digitalization would increase operational efficiencies without increasing market access could lead to job loss as the technology would be more likely to automate jobs than augment them.

Further research argues that public investment can play a role in the spread of digital technologies by providing support for private investment by firms. Fulgenzi⁷¹ argues that budget constraints in sectors like healthcare, which relies heavily on public spending, can lead to underinvestment in digital technologies. Others find that in sectors where the government is a major purchaser of products and services, such as defense, increased government investment will lead to greater technology investment to meet government needs⁷². Economists at the Federal Reserve Bank of Richmond similarly found that sectorally targeted subsidies stimulate technological adoption, particularly in sectors with high ‘multipliers’ where a low fiscal cost of implementation will yield outsized benefits from investment in these subsidies⁷³. These results are intuitive and point to the US government’s key role as a funder of technological advancement through investment in basic research, subsidies and tax credits for strategic industries like energy and technology, and as a major customer of many innovative US companies.

AI Diffusion: Theory and Empirics

Though AI is still only beginning to scale across the US economy, there has been some research into its diffusion, both theoretical and empirical. Theoretical research has attempted to identify some factors which may impact AI’s diffusion, though this research has generally not adopted the sectoral lens used in our framework, instead selecting specific indicators to focus on.

Technical factors at the firm level are a primary consideration in this area of the literature. Dahlke *et al.*⁷⁴ assert that the diffusion of AI will be driven on a firm-by-firm basis based on their level of absorptive capacity, defined as a firm’s ability to recognize and integrate new information and then apply it to its business⁷⁵. Firms with the technical skills, tools, and processes in place to understand and utilize AI are, intuitively, more likely to adopt it and realize value from its implementation⁷⁶. Extrapolating this to the sectoral level, it is likely that sectors with better technological foundations and digitally literate workforces will therefore more rapidly adopt AI.

Other research explores the impact of the economic and regulatory environment firms occupy on their likelihood of adopting AI and other digital technologies. Agrawal, Gans and Goldfarb⁷⁷ argue that subsidies and intellectual property law, key influencing factors of diffusion of other digital

technologies⁷⁸, will have large effects on the adoption of AI. Additionally, they assert that three specific factors: privacy regulations, liability laws, and trade policy, may be more impactful to AI's diffusion than they have been to the spread of previous digital technologies. They also contend that labor laws and competition policy will be key drivers of policy responses to AI's spread, and may impact diffusion if workers or competition authorities are able to use their power to stop the spread of these technologies. Researchers at Brookings argue similarly that labor power will be a critical factor in the diffusion of AI⁷⁹. Drawing upon research into economic exposure to AI, they assert that the roles which may be most susceptible to augmentation or automation from increasingly powerful AI systems are generally those where union membership is the lowest.

In addition to theoretical work, some early research attempts to track the adoption of AI, providing valuable insights regarding the factors which may impact adoption most significantly. Numerous consultancies and researchers have attempted to track where AI has been adopted across industries thus far^{80,81}, but fewer have developed clear frameworks for understanding the factors driving how AI may be adopted across firms of different industries. McElheran *et al.*⁸² posit that while AI use has been found in every sector of the economy, its adoption thus far is the highest in sectors that have already adopted complementary technologies such as cloud computing and robotics. Additionally, AI diffusion has so far been most widespread in higher growth sectors with high levels of venture capital funding. AI's spread in the US so far has been geographically limited to "superstar cities" where many high-growth firms are located⁸³, in line so far with predictions that job exposure to AI will be highest in urban areas due to the density of high tech and white collar work⁸⁴.

As the diffusion of AI continues, additional factors impacting which sectors will most rapidly adopt AI and how pervasive AI becomes within those sectors will become clearer. As this diffusion occurs, it is useful to have an organizing framework to systematically consider the factors impacting how quickly and broadly AI is adopted. Our analysis creates such a framework, consolidating these factors in a way that can be used to determine the potential impact of AI to individual sectors based on their unique characteristics.

Methodology

To conduct our research, we first performed an extensive survey of existing literature on transformative AI and its economic consequences. The goal of this review was to understand the current level of discussion of how transformative AI may diffuse across the economy. We found that the vast majority of related research was focused on the potential for AI to automate or augment various types of tasks, often skipping over the structural, technological, and cultural factors that may increase friction around AI adoption within different sectors.

After surveying the existing TAI literature, we began to map out the gaps that we believed were present in current discussions of AI diffusion. We found ample research which approaches this question from the bottom up, using task-based models and projections from AI experts to predict the types of tasks and associated job roles which may be augmented or automated by AI. But higher level, sectoral factors are often missing from these discussions. We began building an intuitive list of sectoral factors which were consistently omitted in the existing body of bottom-up AI diffusion research but could theoretically impact diffusion. These factors encompassed a wide variety of metrics such as levels and sources of investment, human capital and digital literacy, existing technological infrastructure, and regulatory details, which could be measured at the sectoral level. Drawing heavily upon existing empirical and theoretical technology diffusion literature, we looked for evidence to validate the initial assumptions and flesh out the factors which make up our intuitive framework.

We then searched for existing data sources which track these metrics and could therefore be applied to different sectors. Where public data sources were not available for specific metrics, we considered what type of data would be useful to collect, with the hope that future research could begin to track such factors in the future, or, in some cases, that firms or industry experts could leverage private data sources where available to perform sectoral analysis using our framework.

Our goal in accumulating this evidence is twofold:

1. Gather data sources which provide current insights into sectors' likelihood of AI adoption in the short and medium term, and
2. Propose existing or future data sources which can be used over time to track the spread of AI and its impacts across sectors.

After developing each factor from the bottom up, we began to consolidate and group these factors into high-level groups to simplify our model. Together, the categories are intended to provide a clear, memorable structure for analyzing the potential proliferation of AI based on the susceptibility of different industries.

Within each category, we assessed how we expect each factor to contribute to the diffusion of TAI within a sector that more prominently exhibits that factor to determine each factor's directional impact. For example, we expect sectors which are relatively more digitally mature to be more susceptible to faster and more broad diffusion of TAI than other sectors, all else held equal. Today, our framework and factors remain qualitative, though in the future with additional data collection for our framework factors we hope to create a more quantitative scoring system which could demonstrate through statistical analysis not only the directional impact of each factor, but their relative importance as well.

Given the potential complexity of digesting and applying our framework due to the number of categories and factors included, we then created a series of simplified visualizations to better convey our findings and make it easier to apply to industries across the economy. These visualizations are intended to provide a straightforward, memorable distillation of each category and its component factors to help our audience retain our framework and apply it effectively when performing research or making strategic decisions.

Next, to illustrate the way to apply our framework to different industries, we analyzed two sectors, high-frequency trading (HFT) and retail banking, to demonstrate its use. We chose these sectors to demonstrate that this framework can be effectively applied across sectors at different levels of specificity, provided that the data exists. By comparing these sectors, we demonstrate the process of analysis and the insights which can be gleaned from such analysis regarding the likelihood of diffusion of AI across firms and job roles within sectors. The major differences between these two sectors will likely lead to disparate diffusion patterns, demonstrating the likelihood of varied AI adoption even within the same broader industry of financial services. This concrete example shows the importance of approaching AI diffusion with a sectoral perspective to avoid generalizations and better consider the specific ways that TAI will unevenly reshape the economy.

Overview: A Five-Category Framework for Analyzing AI Diffusion Across Industries

Understanding the diffusion of AI, particularly systems with the potential for transformative impact (TAI), across different industries requires a multi-faceted perspective. Assessing mere technological potential proves insufficient; the interplay of technical foundations, human capital dynamics, economic drivers, prevailing market conditions, and established regulatory environments collectively creates vastly different adoption landscapes across sectors. To systematically analyze this complexity, we propose a framework organizing the key determinants of industry-level AI diffusion into five overarching categories:

1. **Technological readiness:** firms' existing technological infrastructure and the availability of AI tools within a given industry
2. **Workforce and human capital:** baseline technological literacy, AI-specific skills, culture, and labor power within given industry
3. **Investment:** inflows of capital from public and private sources and sectoral expenditure on technology by firms
4. **Markets and competition:** demand patterns, industry competitiveness, economic health
5. **Regulation and oversight:** level of regulatory and oversight burden faced by firms within a given industry

Critically, no single category functions in isolation. Technological readiness remains inert without appropriately skilled personnel and necessary investment. Investment decisions are intrinsically shaped by perceived market opportunities and regulatory constraints. Workforce capabilities influence not only the technical feasibility of adoption but also the negotiated terms under which it proceeds. By evaluating these five categories in concert, a more holistic comprehension emerges regarding the complex forces governing AI diffusion. This integrated analysis facilitates a nuanced understanding of why certain industries demonstrate leadership while others lag, and clarifies how targeted interventions might accelerate or strategically guide AI adoption within specific sectoral contexts.

A Five-Category Framework for Analyzing AI Diffusion Across Industries



Technological Readiness Factors

"What infrastructure and tools are available for AI adoption?"

Digital Maturity

Seamless data availability, quality, and integration with existing systems

AI Ecosystem Density

Off-the-shelf frameworks that reduce development barriers

Complementary Technologies

Supporting technologies that enable AI implementation and usage



Workforce & Human Capital Factors

"How prepared is the workforce for AI integration?"

Cultural Resistance to Change

Entrenched cultures that may resist or embrace technological disruption

AI Skills & Expertise

Availability of AI professionals and knowledge transfer mechanisms

Labor Market Dynamics

Workforce availability and organization impacting AI implementation



Investment Factors

"What financial resources are available for AI adoption?"

Private AI Investment

Venture and private equity funding flowing to AI startups

Public AI Investment

Government funding accelerating sectoral AI adoption

AI Capital Expenditures

Corporate spending on AI infrastructure and personnel



Market & Competitive Factors

"How does market structure influence AI adoption?"

Market Competitiveness

Industry rivalry driving innovation and AI adoption

Sector Economic Health

Macroeconomic conditions influencing AI budget and investment

Demand Patterns

Customer demand for AI-enabled products and services



Regulation & Oversight Factors

"How do legal frameworks affect AI deployment?"

Regulatory Burden

Overall complexity and compliance costs of AI implementation

Enforcement & Oversight Strength

Practical application of regulations across jurisdictions

AI-Specific Regulations

Laws and policies directly addressing artificial intelligence

1. Technological Readiness Factors

Technological Readiness Factors "What infrastructure and tools are available for AI adoption?"		
Digital Maturity Seamless data availability, quality, and integration with existing systems KEY INDICATORS ▪ Data Digitization & Quality (digitization rate, metadata standards) ▪ Legacy IT Compatibility (ERP, mainframes) ▪ Data-Sharing & Ecosystems (APIs, data marketplaces) ▪ Tooling & Automation (microservices, CI/CD for models, MLOps pipelines)	AI Ecosystem Density Off-the-shelf frameworks that reduce development barriers KEY INDICATORS ▪ Marketplace of Prebuilt Models (model zoos, Hugging Face) ▪ Domain-Specific LLMs (finance, healthcare, etc.) ▪ Industry-specific AI tools and applications ▪ Availability of specialized APIs and services	Complementary Technologies Supporting technologies that enable AI implementation and usage KEY INDICATORS ▪ Robotics adoption rate/density ▪ IoT device penetration and usage ▪ Cloud infrastructure availability ▪ High-speed network coverage (5G, fiber)

This category represents the fundamental technical infrastructure and capabilities that determine an industry's ability to implement AI solutions effectively. It considers the existing digital foundation and technological investments upon which AI applications can be built, alongside the availability of specialized AI tools and necessary complementary technologies.

As historical diffusion patterns of General Purpose Technologies (GPTs) demonstrate, organizations require the absorptive capacity to integrate and successfully operationalize new technologies. A lack of strong technological foundations creates significant friction, *limiting how quickly and effectively AI can be integrated* and thus slowing the overall rate of diffusion.

Digital Maturity
DEFINITION This factor refers to an industry's overall level of technological sophistication, particularly concerning data infrastructure, IT systems, and digital processes.
EXPLANATION OF IMPACT High digital maturity, characterized by accessible, high-quality data, modern IT systems, and integrated tooling, <i>significantly increases the rate of AI diffusion</i> by lowering adoption barriers. Critically, this encompasses the ease with which an industry's core operational data, knowledge, and workflows can be digitized into formats suitable for massive consumption as AI training data.

For instance, fields where primary outputs are text-based (e.g., coding, many legal analyses) are currently more amenable to rapid AI-driven automation than those reliant on complex, multi-modal sensory inputs (e.g., nuanced UX design iterations) or intricate, real-world physical tasks (e.g., construction, specialized manual assembly). Industries grappling with legacy systems, poor data quality, or data types inherently difficult to capture, structure, and utilize for training advanced AI models face substantial hurdles, *slowing down the rate of diffusion*.

EXAMPLE

The Retail Banking sector often demonstrates high digital maturity with extensive digital records and analytics. In contrast, the Commercial Construction industry, while advancing, may exhibit more varied maturity, particularly in capturing the complex, unstructured data from physical job sites in AI-ready formats.

KEY OBSERVABLE INDICATORS

- Extent of data digitization and quality (including its suitability for AI training)
- Compatibility with legacy IT systems
- Robustness of data-sharing and ecosystems (e.g., APIs, data marketplaces)
- Sophistication of tooling and automation

SAMPLE DATA SOURCES

- Industry benchmarking reports on data management and AI readiness (e.g., Gartner, IDC).
- CIO surveys on IT modernization and technical debt (e.g., KPMG, Deloitte).
- API directories (e.g., ProgrammableWeb); API management platform reports (e.g., Postman).
- Surveys on enterprise software and Robotic Process Automation (RPA) adoption.
- Sector-level digital maturity indices (e.g., updated McKinsey GSI digitization index).

AI Ecosystem Density



DEFINITION

This factor measures the availability, accessibility, and specialization of AI tools, models, and frameworks relevant to a specific industry.

EXPLANATION OF IMPACT

A dense ecosystem, rich with off-the-shelf frameworks and domain-specific models, *accelerates the rate of AI diffusion* by reducing development barriers, costs, and time-to-deployment. The density of this ecosystem influences whether firms primarily "take" existing solutions, "shape" them through customization, or must "make" bespoke AI applications, each path affecting the speed of diffusion.

EXAMPLE

Sectors like Diagnostic Medical Imaging and Pharmaceutical Research (early-stage compound screening) benefit from a growing number of specialized AI tools. In contrast, Automotive Manufacturing, while a significant AI adopter, may need to develop more bespoke AI solutions for its highly specific production environments.

KEY OBSERVABLE INDICATORS

- Marketplaces for pre-built models
- Existence and performance of domain-specific LLMs

SAMPLE DATA SOURCES

- Model repositories (e.g., Hugging Face); Cloud AI platform offerings (e.g., Google, AWS, Azure).
- Academic research databases (e.g., arXiv for specialized models).
- Venture capital investment data in sector-specific AI startups (e.g., PitchBook, CB Insights).
- Industry publications announcing new specialized AI models.

Complementary Technologies



DEFINITION

AI rarely functions in isolation; its impact often depends on integration with other technologies. This factor assesses the availability, maturity, and integration of these complementary technologies (e.g., robotics, IoT devices, cloud computing, high-speed networks) within an industry.

EXPLANATION OF IMPACT

The ready availability and seamless integration of mature complementary technologies *can increase the rate of AI diffusion* by providing necessary data inputs, processing power, or mechanisms for physical action. Conversely, their absence, immaturity, or poor integration acts as a significant bottleneck, *slowing down the rate of AI diffusion* and constraining its potential scope and impact.

EXAMPLE

Advanced AI in Long-Haul Freight Trucking, for autonomous features, heavily depends on high-speed 5G networks and sophisticated sensor suites. Similarly, AI-driven quality control in Large-Scale Food Processing relies on widespread IoT sensor adoption and robust cloud infrastructure.

KEY OBSERVABLE INDICATORS

- Robotics adoption rates
- IoT device penetration
- Cloud infrastructure adoption
- High-speed network availability
- Maturity of integration capabilities

SAMPLE DATA SOURCES

- Robotics adoption reports (e.g., International Federation of Robotics - IFR).
- Market research on IoT adoption by sector (e.g., Statista, IDC).
- Cloud adoption reports (e.g., Flexera "State of the Cloud Report").
- National broadband maps (e.g., FCC National Broadband Map); telecom regulator data.
- Surveys on enterprise integration platform (iPaaS) usage.

2. Workforce and Human Capital Factors

 Workforce & Human Capital Factors "How prepared is the workforce for AI integration?"		
Cultural Resistance to Change Entrenched cultures that may resist or embrace technological disruption KEY INDICATORS - Industry Prestige (perception) - Cultural Norms (tech enthusiasm to risk aversion) - Fast-Follower Pressures (peer pressure to match rivals) - Legacy Hierarchies/Drag (bureaucracy, past tech adoption patterns)	AI Skills & Expertise Availability of AI professionals and knowledge transfer mechanisms KEY INDICATORS - Workforce Competency & Density (concentration of skilled AI professionals) - Educational & Certification Pipeline (STEM and AI-specific program output) - Organizational Knowledge Transfer (upskilling, internal training) - Domain expertise combined with technical AI knowledge	Labor Market Dynamics Workforce availability and organization impacting AI implementation KEY INDICATORS - Industry-specific unemployment rates - Job vacancy rates and unfilled positions - Unionization rates and collective bargaining power - Wage growth trends and labor costs

This category encompasses the social, cultural, and human dimensions shaping AI adoption trajectories. It examines the interplay between organizational cultures, the availability of specialized talent, and labor relations dynamics that can either catalyze or constrain technological transformation. As history shows, technology adoption is not solely determined by technical feasibility or economic logic; it is fundamentally a socio-technical process. Human factors like skills, resistance to change, and labor negotiations heavily influence the speed and nature of diffusion. Understanding these elements is key to assessing how readily, and at what rate, an industry can adapt its workforce and practices to leverage AI.

Cultural Resistance to Change
DEFINITION This factor measures the prevailing sociological and organizational tendencies within an industry—including embedded attitudes, norms, and institutional structures—that either facilitate or impede the adoption of new technologies like AI. EXPLANATION OF IMPACT Deeply embedded cultural norms favoring tradition, hierarchical decision-making, and strong risk aversion can significantly <i>slow down the rate of AI diffusion</i>. Conversely, industries with cultures that value experimentation, embrace rapid change, and exhibit lower risk thresholds tend to adopt AI more quickly, <i>accelerating its diffusion</i>.

EXAMPLE

A highly regulated industry with entrenched professional norms, such as certain traditional legal practices, might exhibit slower AI adoption for core tasks compared to the more rapidly evolving digital marketing sector, which often embraces new analytical tools more readily.

KEY OBSERVABLE INDICATORS

- Perceived impact of AI on an industry's established prestige: the extent to which AI adoption is seen to affirm or challenge the traditional sources of value, respectability, and entrenched status in high-regard fields (e.g., law, specialized consulting).
- Prevailing cultural norms (e.g., tech enthusiasm vs. pronounced risk aversion).
- Fast-follower pressures driven by competitive dynamics.
- Stance and influence of key industry bodies or associations.
- Drag from legacy hierarchies and bureaucratic impediments.

SAMPLE DATA SOURCES

- Industry-specific surveys on organizational culture and attitudes towards innovation.
- Analysis of publications and guidance from influential industry bodies.
- Case studies on technology adoption patterns within specific sectors.
- Reports on corporate governance and risk management practices by industry.

AI Skills & Expertise



DEFINITION

This factor assesses the availability, quality, and distribution of human capital equipped with the necessary expertise to develop, implement, manage, and strategically leverage AI technologies within an industry.

EXPLANATION OF IMPACT

A strong talent pool—comprising data scientists, machine learning engineers, domain experts skilled in applying AI, and AI-literate managers—*increases the rate of AI diffusion*. Conversely, a shortage of

such talent, particularly in specialized industries requiring deep contextual knowledge combined with technical expertise, creates a significant bottleneck that *slows down AI adoption*.

EXAMPLE

The technology sector itself often benefits from a high concentration of AI talent, facilitating rapid innovation. In contrast, industries like primary education may face greater challenges in attracting, retaining, and upskilling specialized AI professionals.

KEY OBSERVABLE INDICATORS

- Workforce competency and density (concentration of AI professionals relative to industry demand).
- Strength of educational and certification pipeline (output of relevant graduates and certified professionals).
- Effectiveness of organizational knowledge transfer systems (e.g., upskilling initiatives, internal training programs, utilization of professional associations for continuous learning).

SAMPLE DATA SOURCES

- Labor market analytics reports on AI talent supply and demand (e.g., from sources like LinkedIn Talent Insights).
- Higher education enrollment and graduation statistics for AI-related disciplines.
- Industry surveys on skills gaps and corporate training investments.
- Membership data and program offerings from professional AI associations.

Labor Market Dynamics



DEFINITION

This factor considers the collective strength, organization, and influence of workers, often through unions or other representative bodies, in shaping the approach to AI adoption within an industry.

EXPLANATION OF IMPACT

Strong labor representation can *slow down the rate of AI diffusion or alter its trajectory*. This may occur as worker representatives negotiate for stricter terms of AI implementation, demand robust job security measures, advocate for comprehensive retraining programs, or insist on

maintaining significant human oversight in AI-augmented processes. Such negotiations and safeguards, while aimed at worker protection, can extend AI deployment timelines compared to sectors with less collective worker influence.

EXAMPLE

Highly unionized sectors like public transportation or certain manufacturing segments might experience more phased and negotiated AI rollouts that emphasize augmentation, compared to less unionized sectors where automation driven by AI might be introduced more rapidly and with fewer constraints.

KEY OBSERVABLE INDICATORS

- Unionization rates within the industry.
- Content and prevalence of collective bargaining agreements specifically addressing technological change and AI.
- Legal constraints on layoffs or workforce adjustments (e.g., mandated consultation periods, severance requirements).
- History and outcomes of labor negotiations concerning technology adoption.

SAMPLE DATA SOURCES

- Government labor statistics on union membership by industry (e.g., Bureau of Labor Statistics in the US).
- Academic databases and analyses of collective bargaining agreements.
- Employment law databases and policy papers from organizations like the ILO or OECD.
- Industry reports on labor relations and workforce management practices.

3. Investment Factors

Investment Factors "What financial resources are available for AI adoption?"		
Private AI Investment Venture and private equity funding flowing to AI startups KEY INDICATORS - VC Funding in AI startups and scale-ups - Corporate venture capital activity - Angel investment in early-stage AI companies - M&A Activity (acquisition trends of AI startups)	Public AI Investment Government funding accelerating sectoral AI adoption KEY INDICATORS - Industrial Subsidies / Tax Credits (sector-specific) - R&D Grants (government-funded AI research) - Public procurement of AI technologies - National AI strategies and moonshot projects	AI Capital Expenditures Corporate spending on AI infrastructure and personnel KEY INDICATORS - Hiring for AI Roles - AI & IT Spend (corporate budgets) - Patent Filings (R&D proxy) - Economic Impact & Efficiency (ROI, productivity improvements)

This category addresses the economic landscape and financial dynamics driving AI adoption decisions. It examines how capital flows from private and public sources, alongside company-level expenditure patterns, create distinct environments for technological investment and implementation across industries. Investment acts as the crucial fuel translating potential AI capabilities into deployed solutions; understanding these financial flows and commitments is essential for gauging both the current pace of adoption and the resources available for future diffusion within a sector.

Private AI Investment
DEFINITION This factor measures the flow of non-governmental capital—primarily venture capital (VC), corporate R&D spending, and private equity—into AI development, startups, and implementation initiatives relevant to specific industries. EXPLANATION OF IMPACT Strong private funding <i>accelerates the rate of AI diffusion</i> by fueling innovation, supporting the creation of sector-specific solutions, attracting talent, signaling market confidence, and intensifying competitive pressure for adoption. EXAMPLE The rapid development of AI-powered logistics solutions is heavily driven by VC funding into startups and significant R&D investment by large e-commerce and shipping companies.

KEY OBSERVABLE INDICATORS

- Venture Capital (VC) and Private Equity (PE) funding levels in AI by sector.
- Rate of AI startup creation, funding rounds, and valuation growth.
- Reported corporate R&D expenditure specifically allocated to AI.

SAMPLE DATA SOURCES

- Financial data platforms (e.g., PitchBook, CB Insights, Refinitiv).
- Company annual reports and investor briefings.
- Market research reports on industry-specific R&D and innovation trends.

Public AI Investment



DEFINITION

This factor assesses the role of government funding, subsidies, tax incentives, and grants in directly or indirectly supporting AI research, development, and adoption within industries.

EXPLANATION OF IMPACT

Strategic public investment *can increase the rate of AI diffusion*, particularly by de-risking foundational research, building essential digital infrastructure, complementing private funding in nascent areas, and promoting adoption in sectors aligned with national priorities or where market failures might otherwise lead to underinvestment.

EXAMPLE

Government initiatives funding AI research in healthcare or providing grants for SMEs to adopt AI tools in manufacturing aim to stimulate broader AI integration in those sectors.

KEY OBSERVABLE INDICATORS

- Availability and uptake of industrial subsidies or tax credits for AI adoption/development.
- Funding levels for national and international R&D grants supporting AI research.
- Scale and focus of public AI research initiatives and dedicated funding agencies.

SAMPLE DATA SOURCES

- Government budget allocations and national AI strategy documents.
- Public databases of grant awards (e.g., NIH, NSF in the US; Horizon Europe).
- Reports from international organizations (e.g., OECD) on public R&D spending and innovation policies.

AI Capital Expenditures



DEFINITION

This factor quantifies an industry's direct capital outlay for foundational AI capabilities. It is centrally characterized by **massive investments in specialized AI compute infrastructure** (e.g., large-scale GPU clusters, AI-optimized data centers), alongside substantial spending on AI software platforms, and the strategic **investment in acquiring and developing specialized AI talent**, which constitutes essential human capital.

EXPLANATION OF IMPACT

Significant and targeted AI capital expenditures, particularly for high-capacity compute and the expert personnel to leverage it, *directly and substantially increase the rate of AI diffusion*. These investments build the core operational backbone necessary for developing, deploying, and scaling advanced AI solutions, moving AI from experimental phases to widespread, impactful use.

EXAMPLE

Technology giants and research consortia committing billions to build proprietary generative AI models necessitate enormous CapEx in GPU farms and recruitment of elite AI researchers. Similarly, the automotive industry's pursuit of autonomous driving involves significant CapEx in vehicle sensor suites, simulation platforms, and AI development teams.

KEY OBSERVABLE INDICATORS

- Scale of direct investment in dedicated AI compute infrastructure (e.g., expenditure on GPU clusters, AI-specific data centers, and high-performance edge computing hardware). This is often the largest single AI CapEx component.
- Investment in the acquisition, training, and retention of specialized AI talent (e.g., data scientists, ML engineers, AI systems architects),

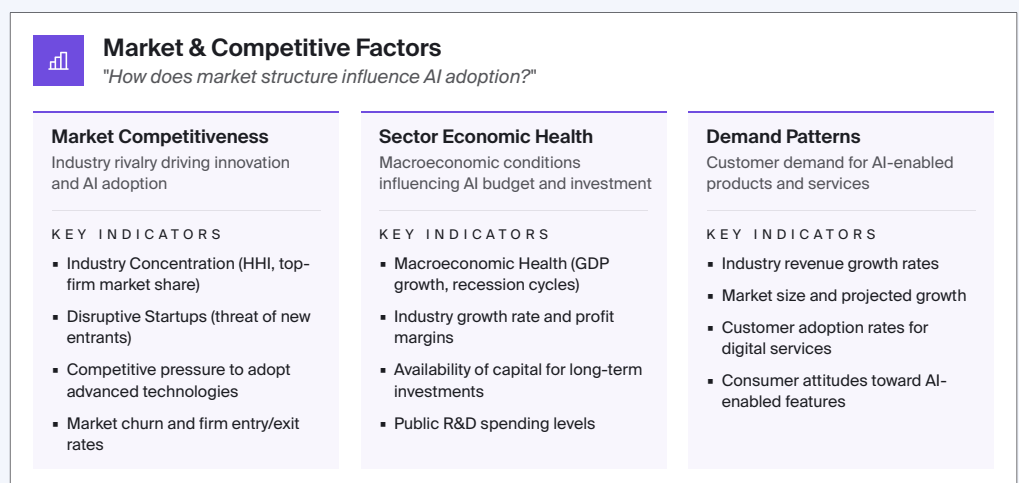
treated as a critical human capital investment to activate and sustain AI capabilities.

- Proportion of corporate IT and operational budgets explicitly allocated to AI platform development, software, and supporting infrastructure.
- AI-related patent filings and substantive M&A activity specifically aimed at acquiring core AI technologies, infrastructure, or established AI teams.
- Quantifiable economic returns and efficiency gains attributed to AI CapEx (e.g., formally tracked ROI, cost reductions from AI-driven automation, and productivity increases linked to AI investments, justifying the expenditure).

SAMPLE DATA SOURCES

- Company financial statements (CapEx disclosures, R&D spending, investor calls detailing AI investments).
- Market research reports on data center, GPU, and enterprise AI spending (e.g., from IDC, Gartner, Omdia).
- Patent office databases (e.g., USPTO, EPO) for AI-specific filings.
- Job market analytics for AI roles and associated compensation benchmarks (reflecting talent investment).
- Industry financial analyst reports and company case studies detailing AI ROI.

4. Market and Competitive Factors



This category focuses on the competitive landscape, prevailing labor market conditions, and underlying demand dynamics within an industry, which collectively shape the primary incentives for AI adoption. Market forces create

the immediate business context, generating pressures and opportunities that strongly influence whether and how organizations invest in AI for efficiency, differentiation, or growth. Understanding these external market drivers is critical for assessing the urgency and strategic rationale behind AI adoption decisions, with robust customer demand often acting as the most potent catalyst among these factors.

Market Competitiveness



DEFINITION

This factor assesses the intensity of rivalry among firms within an industry, including the threat of new entrants and the overall market structure.

EXPLANATION OF IMPACT

Higher market competitiveness generally *increases the rate of AI diffusion*. In intensely competitive environments, firms are often compelled to adopt AI to gain an edge in efficiency, innovate on products/services, or defensively match rivals' capabilities. Conversely, in less competitive, more concentrated markets, the impetus for AI adoption may be slower unless dominant players choose to champion innovation.

EXAMPLE

The global smartphone market, characterized by intense rivalry, sees rapid AI adoption in features like camera enhancements and voice assistants, as firms constantly seek differentiation. In contrast, a more niche, slowly evolving B2B industrial segment with few dominant players might see a slower uptake of AI.

KEY OBSERVABLE INDICATORS

- Measures of industry concentration (e.g., Herfindahl-Hirschman Index - HHI).
- Level of threat from disruptive startups and new market entrants.
- Rates of market churn, firm entry, and exit.
- Intensity of price and non-price (e.g., feature-based) competition.

SAMPLE DATA SOURCES

- Reports from competition authorities and regulatory bodies.
- Market research firm analyses of industry structure and competitive intensity (e.g., IBISWorld, Gartner).

- Startup databases and venture capital investment trends (e.g., PitchBook, Crunchbase).
- Financial news and industry publications detailing competitive actions.

Sector Economic Health



DEFINITION

This factor evaluates the availability of workers relative to employer demand within an industry, which directly influences wage levels and the economic incentives for automation or labor augmentation through AI.

EXPLANATION OF IMPACT

A tight labor market (characterized by low supply and high demand for workers) *tends to accelerate the rate of AI diffusion*. In such conditions, companies face labor shortages and rising wage costs, making AI solutions for task automation or human augmentation more economically attractive and strategically necessary. Conversely, a slack labor market may reduce this urgency.

EXAMPLE

Industries like logistics and warehousing experiencing persistent driver and picker shortages are increasingly adopting AI-powered robotics and route optimization to cope with demand and manage labor costs.

KEY OBSERVABLE INDICATORS

- Industry-specific unemployment rates and labor force participation rates.
- Job vacancy rates and duration of open positions (especially for key operational and technical roles).
- Wage growth trends within the industry.
- Ratios of relevant graduates or skilled individuals to job openings.

SAMPLE DATA SOURCES

- National and regional labor statistics agencies (e.g., Bureau of Labor Statistics in the US, Eurostat).
- Job posting analytics platforms (e.g., Lightcast, LinkedIn Talent Insights).
- Industry-specific salary surveys and HR consulting reports.
- Reports from educational institutions on graduate employment.

Demand Patterns



DEFINITION

This factor measures the level, growth, and nature of customer and end-user demand for an industry's products or services, including any specific appetite for AI-enabled features or improved service levels.

EXPLANATION OF IMPACT

As arguably the most crucial market driver, strong and evolving market demand *powerfully accelerates the rate of AI diffusion*. High or growing demand creates compelling incentives for firms to invest in AI to improve efficiency, scale operations, innovate offerings, personalize experiences, or meet explicit market desires for AI-enhanced products and services. Stagnant or declining demand, conversely, can significantly curb such discretionary investments.

EXAMPLE

The surge in demand for personalized digital entertainment has fueled massive AI investment in recommendation engines and content generation by streaming services. Similarly, increasing consumer expectations for rapid and customized e-commerce experiences drive AI adoption in areas like chatbots, demand forecasting, and personalized marketing.

KEY OBSERVABLE INDICATORS

- Industry revenue growth rates (historical and projected).
- Market size and projected Compound Annual Growth Rate (CAGR).
- Customer adoption rates for new, digital, or AI-enhanced services/products.
- Metrics like order backlogs, capacity utilization, and customer acquisition rates.
- Trends in relevant consumer or business spending patterns impacting the industry.

SAMPLE DATA SOURCES

- Industry market research reports (e.g., from Statista, Gartner, Forrester, Mordor Intelligence).
- Company earnings reports, investor presentations, and public financial filings.
- National economic accounts and consumer spending data from government statistics agencies.
- Surveys on customer preferences and technology adoption.

5. Regulation and Oversight Factors

 Regulation & Oversight Factors "How do legal frameworks affect AI deployment?"		
Regulatory Burden Overall complexity and compliance costs of AI implementation KEY INDICATORS - Compliance cost estimates - Regulatory density indices - Industry/Consortium Standards (ISO, IEEE) - Standards & Interoperability (APIs, data schemas)	Enforcement & Oversight Strength Practical application of regulations across jurisdictions KEY INDICATORS - Regulatory agency budgets and staffing - Frequency and size of penalties - Liability & Risk Framework - Audit requirements and documentation standards	AI-Specific Regulations Laws and policies directly addressing artificial intelligence KEY INDICATORS - AI-Related Statutes (EU AI Act, US Algorithmic Accountability) - Data Protection Regulations (GDPR, HIPAA) - Sector-specific AI regulations (healthcare, finance) - Ethical AI requirements and guidelines

This category examines the formal and informal governance structures—laws, regulations, industry standards, enforcement mechanisms, and liability frameworks—that shape the boundaries, requirements, and accountability for AI deployment across industries. The regulatory environment establishes the "rules of the game" for AI innovation and implementation, significantly impacting risk assessment, compliance costs, investment certainty, and ultimately, the design and deployment choices made by firms. A clear understanding of these regulatory contours is essential, as they can act as either significant accelerators (e.g., through clear, enabling standards) or potent inhibitors of AI diffusion.

Regulatory Burden
DEFINITION This factor assesses the overall complexity, stringency, and cost implications of the existing regulatory landscape (excluding AI-specific laws, which are covered separately) that affects AI development and deployment within an industry.
EXPLANATION OF IMPACT A high regulatory burden, characterized by complex, ambiguous, or stringent regulations and a lack of clear, supportive standards, generally <i>slows down the rate of AI diffusion</i>. This is due to increased compliance costs, heightened uncertainty, and potential delays in implementation. Conversely, while the overall burden is a negative factor, the presence of well-defined, widely adopted technical standards and interoperability

frameworks can *mitigate this burden* by reducing friction, enhancing predictability, and building trust, thereby facilitating smoother AI integration.

EXAMPLE

The pharmaceutical industry, with its stringent safety and efficacy regulations, may see a more deliberate and costly AI adoption path for core R&D and clinical trial processes compared to less regulated sectors like online entertainment, where AI tools can be deployed more rapidly.

KEY OBSERVABLE INDICATORS

- Presence and maturity of enabling standards and interoperability frameworks (e.g., from ISO, IEEE) that facilitate AI system integration and data exchange.
- Overall regulatory intensity (e.g., general manufacturing, environmental, labor laws) that AI systems must nonetheless accommodate.
- Compliance cost estimates or sector-specific regulatory impact analyses.

SAMPLE DATA SOURCES

- Reports from standards development organizations (e.g., ISO, IEEE) and industry consortia.
- Regulatory impact assessments and compliance cost studies by industry.
- Surveys of businesses on perceived regulatory challenges and administrative burdens.
- Legal and consulting firm analyses of sectoral regulatory landscapes.

Enforcement and Oversight Strength



DEFINITION

This factor evaluates the nature, rigor, and predictability of monitoring, enforcement actions, and liability frameworks associated with the regulatory environment relevant to AI in an industry.

EXPLANATION OF IMPACT

While predictable and clear enforcement of well-understood rules can foster trust, "Enforcement and Oversight Strength" is considered a negative factor here because increasing intensity (e.g., high scrutiny, severe penalties, complex oversight mechanisms) generally *slows the rate of AI diffusion* by heightening perceived risks and compliance efforts. Weak, inconsistent, or unpredictable oversight can also deter adoption by creating an unstable environment and unclear accountability, whereas overly aggressive or unclear enforcement can stifle innovation due to fear of non-compliance.

EXAMPLE

In financial services, rigorous enforcement of data security and algorithmic trading regulations, while necessary, requires significant investment in AI governance and compliance systems, potentially moderating the speed of new AI deployments. Conversely, sectors with historically lax oversight might see haphazard AI adoption until a major incident triggers stronger enforcement.

KEY OBSERVABLE INDICATORS

- Enforcement vigor: Resources of regulatory agencies, frequency and magnitude of penalties for non-compliance, stringency of audit and reporting requirements.
- Regulatory clarity and consistency: Degree of ambiguity in regulatory interpretation or inconsistency in enforcement across jurisdictions or over time.
- Liability and risk allocation frameworks: Clarity of laws defining liability for AI system failures, availability and cost of relevant insurance, and prevalence of mandated risk assessment or accountability frameworks.

SAMPLE DATA SOURCES

- Annual reports and enforcement statistics from regulatory agencies.
- Legal databases and analyses of case law related to regulatory breaches.
- Reports on AI risk management and insurance products for AI liabilities.
- Studies on corporate compliance practices and perceived enforcement risks.

AI-Specific Regulations



DEFINITION

This factor focuses on the existence, scope, clarity, and prescriptive nature of laws, regulations, and formal ethical guidelines explicitly targeting the development, deployment, and use of AI technologies.

EXPLANATION OF IMPACT

The introduction of new AI-specific regulations, while often aimed at ensuring safety, fairness, and transparency, typically *slows down the rate of AI diffusion, at least initially*. This is because firms must invest in understanding and complying with new requirements, adapting system designs, and implementing governance processes, which can add costs and extend development timelines. This effect is compounded by adjacent regulations crucial for AI.

EXAMPLE

The EU AI Act imposes varying requirements based on risk levels, directly shaping how AI systems are developed and deployed within the EU market. Similarly, sector-specific FDA guidelines for AI in medical devices dictate a rigorous validation and approval process.

KEY OBSERVABLE INDICATORS

- Existence and maturity of horizontal AI laws (e.g., comprehensive AI acts like the EU AI Act, national algorithmic accountability laws).
- Data governance and privacy regulations impacting AI: Strictness and applicability of data protection laws (e.g., GDPR, CCPA, HIPAA) which are foundational for data-intensive AI.
- Sector-specific AI regulations and standards: Rules governing AI use in particular domains demanding high levels of safety, ethics, or scrutiny (e.g., finance, healthcare, autonomous vehicles, public sector).

SAMPLE DATA SOURCES

- Official texts of AI-specific legislation and regulations (e.g., EU AI Act, national AI strategies).
- Guidance documents from data protection authorities (DPAs).
- Publications and standards from sector-specific regulatory bodies (e.g., FDA, EASA, financial conduct authorities).
- Legal analyses and white papers on AI regulatory compliance.

Framework Demonstration: High-Frequency Trading and Retail Banking Sectors

High-Frequency Trading: Overview

High-frequency trading (HFT) is positioned for rapid and substantial TAI diffusion due to its unique industry characteristics. The sector is likely to experience accelerated adoption of AI with comprehensive transformation occurring significantly faster than most financial sub-sectors. HFT's data-centric and algorithmically complex foundations coupled with intense competitive pressures to discover and exploit marginal performance advantages make it susceptible to such widespread and accelerated impacts from TAI.

TAI is likely to automate the majority of labor in the HFT sector while augmenting a smaller percentage of remaining human roles. The industry has already automated substantial execution components, with humans primarily improving trading algorithms, overseeing risk parameters, and guiding strategic decisions. TAI will accelerate this trend with increasingly autonomous trading systems capable of:

- Automating strategy development through automated backtesting and parameter optimization
- Developing novel trading approaches through reinforcement learning
- Self-adjusting to changing market conditions without human intervention

However, regulators will likely maintain requirements for human oversight of risk limits and ultimate algorithmic accountability, preserving critical augmentation components. Humans may also remain in managerial roles, leading teams of AI agents to monitor performance and adjust high-level trading strategies.

The sector represents one of the clearest cases for substantial TAI diffusion within financial services. Extraordinary competitive pressures, large amounts of private capital available for investment, low labor power, and high technological readiness and workforce skills create optimal conditions for widespread adoption of TAI.

High-Frequency Trading: Key Diffusion Enablers

- 1. Exceptional Digital Maturity:** HFT firms already operate with cutting-edge data infrastructure, with near-complete digitization of trading processes and exceptional data quality requirements. The industry maintains some of the most sophisticated IT systems worldwide, with significant investments in low-latency communications technologies and customized high-performance computing systems.
- 2. Strong Private AI Investment:** Large quantitative trading firms are extremely wealthy and are consistently willing to spend top dollar to attract top-level talent and invest in leading-edge technology, demonstrating substantial financial commitment to gaining competitive advantages.
- 3. Extreme Market Competitiveness:** The hypercompetitive nature of HFT creates enormous pressure for technological advancement. With profit margins increasingly compressed, firms face existential pressure to adopt cutting-edge AI solutions to research and develop new trading strategies and deliver microsecond advantages in trading execution or decision-making.
- 4. High AI Skills Availability:** While there is a limited pool of experts at the intersection of quantitative finance and advanced AI knowledge, top HFT firms often attract technical talent from academia and other industries with large pay packages. This will allow for the increased development and integration of increasingly powerful AI systems into firms' trading systems.

High-Frequency Trading: Primary Diffusion Bottleneck

- 1. Explainability Challenges in Regulatory Compliance:** As regulatory oversight intensifies, HFT firms face growing pressure to demonstrate explainability and fairness in algorithmic trading strategies. The SEC's enforcement actions against firms deploying AI algorithms in ways deemed unsafe demonstrate a desire to limit the spread of AI algorithms⁸⁵.

Key Factors Influencing HFT Diffusion



Technological Readiness Factors

"What infrastructure and tools are available for AI adoption?"

Digital Maturity

Seamless data availability, quality, and integration with existing systems

AI Ecosystem Density

Off-the-shelf frameworks that reduce development barriers

Complementary Technologies

Supporting technologies that enable AI implementation and usage



Workforce & Human Capital Factors

"How prepared is the workforce for AI integration?"

Cultural Resistance to Change

Entrenched cultures that may resist or embrace technological disruption

AI Skills & Expertise

Availability of AI professionals and knowledge transfer mechanisms

Labor Market Dynamics

Workforce availability and organization impacting AI implementation



Investment Factors

"What financial resources are available for AI adoption?"

Private AI Investment

Venture and private equity funding flowing to AI startups

Public AI Investment

Government funding accelerating sectoral AI adoption

AI Capital Expenditures

Corporate spending on AI infrastructure and personnel



Market & Competitive Factors

"How does market structure influence AI adoption?"

Market Competitiveness

Industry rivalry driving innovation and AI adoption

Sector Economic Health

Macroeconomic conditions influencing AI budget and investment

Demand Patterns

Customer demand for AI-enabled products and services



Regulation & Oversight Factors

"How do legal frameworks affect AI deployment?"

Regulatory Burden

Overall complexity and compliance costs of AI implementation

Enforcement & Oversight Strength

Practical application of regulations across jurisdictions

AI-Specific Regulations

Laws and policies directly addressing artificial intelligence

Retail Banking: Overview

Retail banking is positioned for **moderate but substantial TAI diffusion**. The diffusion pattern will likely be uneven, with customer-facing applications advancing rapidly while complex legacy core banking systems may limit the potential for a radical technological overhaul of the sector.

The sector's complex customer relationships, regulatory requirements, and trust dynamics favor human-AI collaboration over pure automation approaches. Therefore, we project that TAI will **augment most retail banking jobs**, particularly customer-facing roles, while automating most back-office operational tasks. Customers' desire for increasingly personalized banking experiences will likely drive retail banks to deploy AI systems to better serve client needs, though these experiences may continue to be intermediated through human advisors.

Key dynamics include:

- Customer-facing roles evolving toward augmented advisory capabilities, with AI handling data analysis to improve personalization while humans manage relationship complexity

- Middle-office functions seeing significant automation of routine compliance monitoring, documentation review, and risk assessment
- Back-office operations such as customer service experiencing the highest and most rapid automation rates through intelligent process automation⁸⁶
- Core banking applications demonstrating stickiness due to their complexity and the risk and cost to re-engineer them

Overall, TAI diffusion in retail banking will be significant, but will likely focus on tactical efficiency and customer experience improvements rather than completely reshaping the industry. Key services like fraud prevention, KYC, and customer service will continue to be automated and digital banking platforms will be enhanced using more complex AI algorithms. Customer demand for human interactions for advisory services will likely preserve many human roles, even in cases where AI systems may be more efficient from an economic perspective.

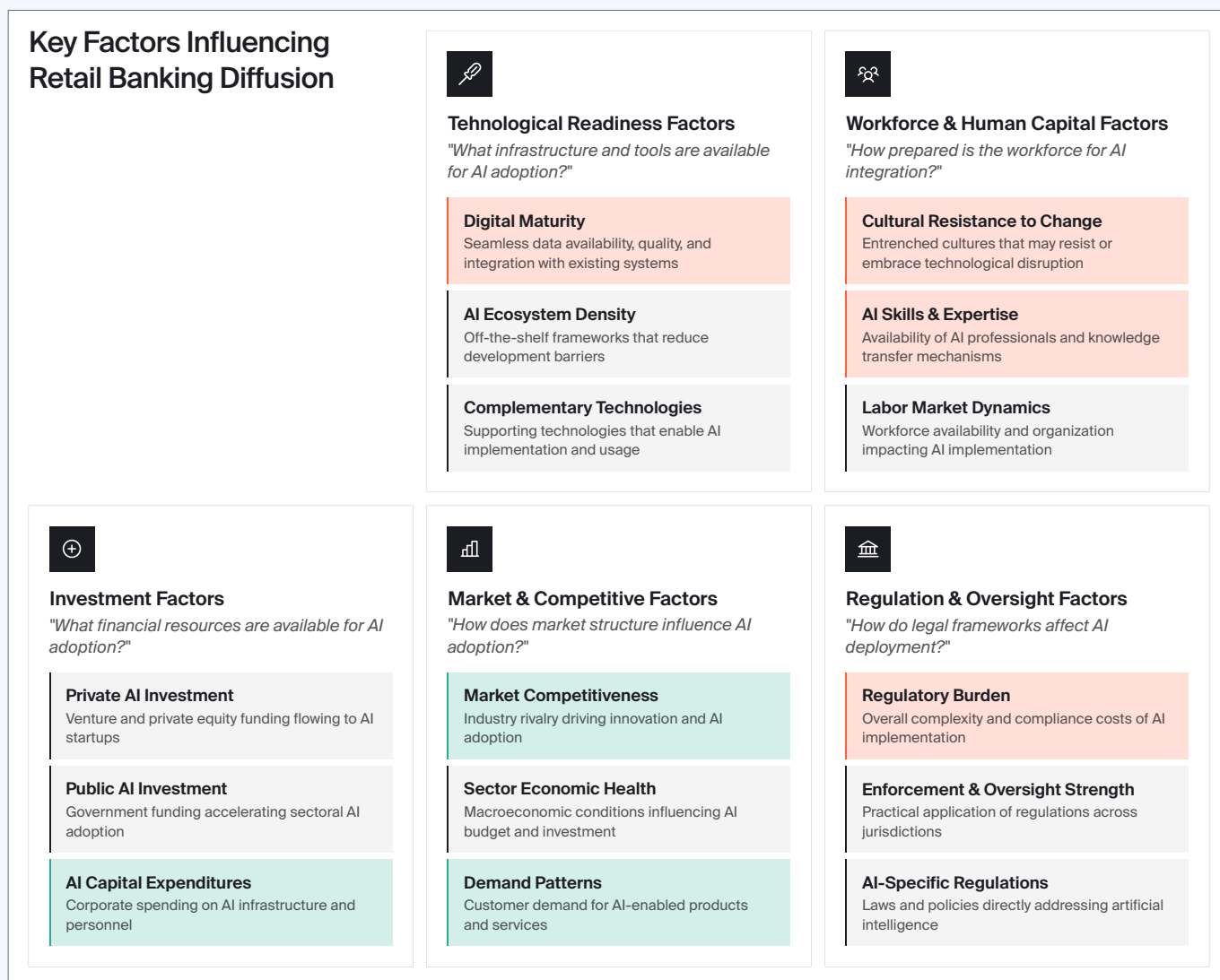
Retail Banking: Key Diffusion Enablers

1. **Strong Demand Drivers:** Consumer expectations increasingly favor digital-first, personalized banking experiences. This will drive banks to invest heavily in AI to drive improved data analysis and recommendation algorithms to deliver tailored solutions for individual customers.
2. **Substantial AI Capital Expenditures:** The banking sector, historically a large IT spender, is investing heavily in developing and acquiring AI solutions to attempt to capture the value that AI may unlock within banking.

Retail Banking: Primary Diffusion Bottlenecks

1. **Legacy IT Systems:** The banking industry's digital maturity is hampered by extensive legacy infrastructure, creating integration challenges and technical debt which may limit TAI's transformative potential in the sector.
2. **Regulatory Burden:** The highly regulated nature of banking creates substantial friction for AI implementation. Strict risk management controls, KYC requirements, and explainability standards will add complexity to AI deployments and could slow adoption.
3. **Cultural Resistance to Change:** Traditional banking cultures, particularly in mid-size and regional institutions, exhibit significant risk aversion. This may create an uneven level of diffusion where larger national banks deploy AI solutions more rapidly while regional institutions lag behind, mirroring current and historical technology adoption trends in the sector⁸⁷.

4. AI Skills and Expertise: While the broader financial sector has strong technical skills, much of this talent is deployed within higher margin areas of the banking business rather than retail banking. This may limit the initial integration of AI applications within retail banking, though increasingly powerful AI systems could help to bridge this skills gap.



Comparative Analysis: High-Frequency Trading vs Retail Banking

Overall Susceptibility Assessment

High-frequency trading demonstrates **exceptional susceptibility** to TAI diffusion, with projected comprehensive transformation within 2-3 years after the development of TAI. The sector's high digitization levels and data quality, algorithmic foundations, and fierce competitive dynamics create an optimal environment for rapid, deep AI integration. TAI will likely reshape HFT's competitive landscape fundamentally, with early adopters gaining substantial advantages in strategy development, execution capabilities, and risk management. Possible regulatory changes represent a potential obstacle for rapid TAI rollout, as financial markets overseers have demonstrated a desire for some level of transparency in algorithmic trading.

In contrast, retail banking exhibits **moderate susceptibility** with a protracted 5-7 year transformation timeline post-TAI. The sector's diffusion pattern will be markedly uneven, progressing through distinct phases: customer-facing applications first (e.g., chatbots, personalized recommendations), followed by middle-office functions (e.g., fraud detection, compliance), and finally core banking operations. This staggered adoption reflects the sector's complex legacy constraints, regulatory burdens, limited technical skills, and organizational inertia. Further discrepancies may emerge between large leading banks with more talent and capital resources and regional banks which lag technologically and prioritize human interactions over advanced digital banking capabilities.

Key Implications for TAI Diffusion

- 1. Diffusion Speed Divergence:** The projected 3-4 year gap in comprehensive transformation timelines between these sectors does not reflect differences in TAI's potential value, but rather the substantial variation in adoption prerequisites, implementation complexity, and organizational readiness.
- 2. Implementation Approach Distinction:** HFT will likely pursue revolutionary adoption strategies with comprehensive system replacements and fundamental strategy reimagining, while retail banking will follow evolutionary paths with gradual capability enhancement and

phased implementation across functional areas.

- 3. Economic Impact Variation:** The concentrated nature of HFT means TAI diffusion will yield intense but narrow economic effects, primarily restructuring market microstructure and trading firm profitability. Banking's broader economic integration means its slower transformation will ultimately generate more extensive societal impacts through changed consumer financial experiences, credit accessibility, and financial inclusion dynamics.
- 4. Capability Development Trajectory:** HFT's advanced technological foundation means firms can immediately focus on developing and deploying sophisticated TAI applications, while banks must initially invest in foundational capability building (improving data architecture, integration layers, and modernization efforts) before developing advanced applications or adopting and integrating off-the-shelf applications.
- 5. Competitive Landscape Restructuring:** TAI adoption will likely accelerate consolidation in HFT as technological advantages compound, while potentially introducing more fragmentation in banking by further lowering barriers to entry for fintech competitors specializing in specific banking functions enhanced by AI capabilities.

The stark contrast between these financial subsectors underscores how industry characteristics fundamentally shape technology diffusion patterns. HFT represents an ideal case for rapid, comprehensive TAI adoption due to its digital-native structure, competitive intensity, and alignment between AI capabilities and core value drivers. Retail banking's more measured transformation reflects the complex reality of TAI diffusion in sectors with substantial legacy constraints, regulatory complexity, and organizational scale and inertia.

Conclusion

The diffusion of Transformative AI (TAI), far from being an instantaneous or uniform event, is likely to be a complex and uneven process shaped by a confluence of technological, cultural, and structural factors that vary by industry. Forecasting the pace and scale of TAI diffusion is a critical endeavor not only to predict the ultimate balance of human and AI labor across sectors of the economy, but to better understand where TAI will have the most immediate impacts. Such insights can better prepare workers, business leaders, and policymakers to adapt to a post-TAI economy. To this end, our organizing framework provides a holistic lens to structure future discussions of the breadth and speed of TAI diffusion.

Our qualitative, sectoral diffusion framework is intended to be a starting point for further analysis of TAI's proliferation across the economy. We envision multiple paths for future research into TAI diffusion:

1. **Econometric analysis** of factors' impacts on AI diffusion over time to develop weights for individual factors, and improve the predictive power of our framework going forward.
2. Expanding beyond our sectoral scope to develop a **macro-level geographic framework** to assess the ways that foundational social, economic, cultural, and institutional factors will impact TAI diffusion across regions.
3. Creating a **composite diffusion index**, leveraging sectoral, geographic, and task-based data to forecast and track the overall societal level of AI adoption.

Each of these future research paths requires **improved data collection** over time, both for the observable indicators underpinning each of our framework factors and for the level of AI diffusion across the economy. Today, much government labor market data, business survey and earnings data, and third-party aggregate data is published annually or quarterly, making it difficult to build an up-to-date view of the diffusion of AI. As we approach TAI, it is reasonable to predict that diffusion speed will increase, making **data lags a critical challenge** to understanding the current level and rate of change of AI adoption metrics.

Improved data collection and transparency by governments, businesses, and third-parties will greatly improve our collective ability to **track diffusion patterns** over time. We urge policymakers and researchers to advance these data collection efforts, which will better enable us to anticipate labor market shifts, guide AI governance initiatives, and design targeted interventions in a period of intense technological transformation.

References

- 1 Qureshi, Z. (2025) *Pathways to short TAI timelines*. Convergence Analysis. Available at: <https://www.convergenceanalysis.org/research/pathways-to-short-tai-timelines> (Accessed: 27 February 2025).
- 2 Cheng, D. et al. (2025) *Threshold 2030: Modeling AI Economic Futures*. Conference Report. Convergence Analysis. Available at: <https://www.convergenceanalysis.org/threshold-2030/> (Accessed: 25 February 2025).
- 3 Wynroe, K., Atkinson, D. and Sevilla, J. (2023) *Literature Review of Transformative Artificial Intelligence Timelines*. Epoch AI. Available at: <https://epoch.ai/blog/literature-review-of-transformative-artificial-intelligence-timelines> (Accessed: 27 February 2025).
- 4 Edwards, B. (2025) *Anthropic chief says AI could surpass “almost all humans at almost everything” shortly after 2027*, *Ars Technica*. Available at: <https://arstechnica.com/ai/2025/01/anthropic-chief-says-ai-could-surpass-almost-all-humans-at-almost-everything-shortly-after-2027/> (Accessed: 14 May 2025).
- 5 Pillay, T. (2025) *How OpenAI’s Sam Altman Is Thinking About AGI and Superintelligence in 2025*, *TIME*. Available at: <https://time.com/7205596/sam-altman-superintelligence-agi/> (Accessed: 14 May 2025).
- 6 Dizikes, P. (2024) *Daron Acemoglu: What do we know about the economics of AI?*, *MIT Economics*. Available at: <https://economics.mit.edu/news/daron-acemoglu-what-do-we-know-about-economics-ai> (Accessed: 14 May 2025).
- 7 Acemoglu, D. and Johnson, S. (2024) ‘Learning From Ricardo and Thompson: Machinery and Labor in the Early Industrial Revolution and in the Age of Artificial Intelligence’, *Annual Review of Economics*, 16(Volume 16, 2024), pp. 597–621. Available at: <https://doi.org/10.1146/annurev-economics-091823-025129>.
- 8 Deming, D.J., Ong, C. and Summers, L.H. (2025) ‘Technological Disruptions in the Labor Market’. Cambridge, MA (NBER Working Paper Series, 33323). Available at: <https://www.nber.org/papers/w33323> (Accessed: 1 March 2025).
- 9 Aghion, P., Jones, B.F. and Jones, C.I. (2018) ‘Artificial Intelligence and Economic Growth’, in *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, pp. 237–282. Available at: <https://www.nber.org/books-and-chapters/economics-artificial-intelligence-agenda/artificial-intelligence-and-economic-growth> (Accessed: 14 May 2025).
- 10 Brynjolfsson, E. (1993) ‘The productivity paradox of information technology’, *Commun. ACM*, 36(12), pp. 66–77. Available at: <https://doi.org/10.1145/163298.163309>.
- 11 Helpman, E. and Trajtenberg, M. (1996) *Diffusion of General Purpose Technologies*. w5773. Cambridge, MA: National Bureau of Economic Research, p. w5773. Available at: <https://doi.org/10.3386/w5773>.
- 12 Hu, K. and Hu, K. (2023) ‘ChatGPT sets record for fastest-growing user base - analyst note’, *Reuters*, 2 February. Available at: <https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/> (Accessed: 22 April 2025).
- 13 Acemoglu, D. and Restrepo, P. (2018) *Artificial Intelligence, Automation and Work*. w24196. Cambridge, MA: National Bureau of Economic Research, p. w24196. Available at: <https://doi.org/10.3386/w24196>.
- 14 Acemoglu, D., Kong, F. and Restrepo, P. (2024) ‘Tasks at Work: Comparative Advantage, Technology and Labor Demand’. Available at: <https://www.nber.org/papers/w32872> (Accessed: 25 February 2024).
- 15 Ha, 2025
- 16 Kelly, J. (2025) *Companies Are Replacing Workers With AI—Fast*, *Forbes*. Available at: <https://www.forbes.com/sites/jackkelly/2025/05/04/its-time-to-get-concerned-klarna-ups-duolingo-cisco-and-many-other-companies-are-replacing-workers-with-ai/> (Accessed: 13 May 2025).
- 17 Villalobos, P. (2023) *Scaling Laws Literature Review*, *Epoch AI*. Available at: <https://epoch.ai/blog/scaling-laws-literature-review> (Accessed: 3 April 2025).
- 18 Cheng et al., 2025 (see n. 2)
- 19 Sevilla, J. (2024) *Can AI Scaling Continue Through 2030?*, *Epoch AI*. Available at: <https://epoch.ai/blog/can-ai-scaling-continue-through-2030> (Accessed: 3 April 2025).
- 20 Wynroe, Atkinson and Sevilla, 2023 (see n. 3)
- 21 McCarthy, J. (2007) ‘From here to human-level AI’, *Artificial Intelligence*, 171(18), pp. 1174–1182. Available at: <https://doi.org/10.1016/j.artint.2007.10.009>.
- 22 Hodson, H. (2019) ‘DeepMind and Google: the battle to control artificial intelligence’, *The Economist*, 1 March. Available at: <https://www.economist.com/1843/2019/03/01/deepmind-and-google-the-battle-to-control-artificial-intelligence> (Accessed: 3 April 2025).
- 23 Baum, J. and Villasenor, J. (2023) *How close are we to AI that surpasses human intelligence?*, *Brookings*. Available at: <https://www.brookings.edu/articles/how-close-are-we-to-ai-that-surpasses-human-intelligence/> (Accessed: 3 April 2025).

- 24 Qureshi, 2025 (see n. 1)
- 25 Gruetzemacher, R. and Whittlestone, J. (2022) 'The transformative potential of artificial intelligence', *Futures*, 135, p. 102884. Available at: <https://doi.org/10.1016/j.futures.2021.102884>.
- 26 Qureshi, 2025 (see n. 1)
- 27 Karnofsky, H. (2016) 'Some Background on Our Views Regarding Advanced Artificial Intelligence', *Open Philanthropy*, 6 May. Available at: <https://www.openphilanthropy.org/research/some-background-on-our-views-regarding-advanced-artificial-intelligence/> (Accessed: 22 May 2025).
- 28 Zhang, B. and Dafoe, A. (2019) 'Artificial Intelligence: American Attitudes and Trends'. Rochester, NY: Social Science Research Network. Available at: <https://doi.org/10.2139/ssrn.3312874>.
- 29 Dafoe, A. (2018) *AI Governance: A Research Agenda*. Centre for the Governance of AI. Available at: <https://www.fhi.ox.ac.uk/wp-content/uploads/GovAI-Agenda.pdf> (Accessed: 1 April 2025).
- 30 Deming, Ong and Summers, 2025 (see n. 8)
- 31 West, D.M. and Allen, J.R. (2018) *How artificial intelligence is transforming the world*. Brookings Institution. Available at: <https://www.brookings.edu/articles/how-artificial-intelligence-is-transforming-the-world/> (Accessed: 16 March 2025).
- 32 McKinsey & Company (2024) *The state of AI in early 2024*, McKinsey & Company. Available at: <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai> (Accessed: 25 February 2025).
- 33 Handa, K. et al. (2025) 'Which Economic Tasks are Performed with AI? Evidence from Millions of Claude Conversations'. Available at: <https://www.anthropic.com/news/the-anthropic-economic-index> (Accessed: 8 March 2025).
- 34 Acemoglu, Kong and Restrepo, 2024 (see n. 14)
- 35 Korinek, A. and Juelfs, M. (2022) 'Preparing for the (non-existent?) future of work'. Washington, DC (Brookings Center on Regulations and Markets). Available at: <https://www.brookings.edu/articles/preparing-for-the-non-existent-future-of-work/> (Accessed: 27 February 2025).
- 36 Korinek, A. (2024) *The Economics of Transformative AI*, NBER. Available at: <https://www.nber.org/reporter/2024number4/economics-transformative-ai> (Accessed: 28 February 2025).
- 37 Korinek, 2024 (see n. 35)
- 38 Goel, S. and Kovács-Ondrejko, O. (2023) *Reskilling for a Rapidly Changing World*, BCG Global. Available at: <https://www.bcg.com/publications/2023/reskilling-workforce-for-future> (Accessed: 25 February 2025).
- 39 Owen, D. and Besiroglu, T. (2023) *Challenges in Predicting AI Automation*, Epoch AI. Available at: <https://epoch.ai/blog/challenges-in-predicting-ai-automation> (Accessed: 9 February 2025).
- 40 Korinek, A. and Suh, D. (2024) 'Scenarios for the Transition to AGI'. arXiv. Available at: <https://doi.org/10.48550/arXiv.2403.12107>.
- 41 Acemoglu and Restrepo, 2018 (see n. 13)
- 42 Qureshi, 2025 (see n. 1)
- 43 Korinek, 2024 (see n. 35)
- 44 Georgieva, K. (2024) *AI Will Transform the Global Economy. Let's Make Sure It Benefits Humanity*, IMF. Available at: <https://www.imf.org/en/Blogs/Articles/2024/01/14/ai-will-transform-the-global-economy-lets-make-sure-it-benefits-humanity> (Accessed: 3 April 2025).
- 45 Frey, C.B. and Osborne, M.A. (2017) 'The future of employment: How susceptible are jobs to computerisation?', *Technological Forecasting and Social Change*, 114, pp. 254–280. Available at: <https://doi.org/10.1016/j.techfore.2016.08.019>.
- 46 Pizzinelli, C. et al. (2023) *Labor Market Exposure to AI: Cross-country Differences and Distributional Implications*, IMF. Available at: <https://www.imf.org/en/Publications/WP/Issues/2023/10/04/Labor-Market-Exposure-to-AI-Cross-country-Differences-and-Distributional-Implications-539656> (Accessed: 3 April 2025).
- 47 Sharps, S. et al. (2024) *The Impact of AI on the Labour Market*. Tony Blair Institute. Available at: <https://institute.global/insights/economic-prosperity/the-impact-of-ai-on-the-labour-market> (Accessed: 14 May 2025).
- 48 Gmyrek, P., Berg, J. and Bescond, D. (2023) *Generative AI and jobs: a global analysis of potential effects on job quantity and quality*. Geneva, Switzerland: International Labour Organization (ILO working paper / International Labour Organization, 96 (August 2023)). Available at: <https://doi.org/10.54394/FHEM8239>.
- 49 Korinek, A. and Stiglitz, J.E. (2017) 'Artificial Intelligence and its Implications for Income Distribution and Unemployment'. Cambridge, MA (NBER Working Paper Series, 24174). Available at: <http://www.nber.org/papers/w24174> (Accessed: 14 May 2025).
- 50 Frank, M.R., Ahn, Y.-Y. and Moro, E. (2025) 'AI exposure predicts unemployment risk: A new approach to technology-driven job loss', *PNAS Nexus*, 4(4), p. pgaf107. Available at: <https://doi.org/10.1093/pnasnexus/pgaf107>.
- 51 Cheng et al., 2025 (see n. 2)
- 52 Korinek and Stiglitz, 2017 (see n. 48)
- 53 Davidson, T. (2023) 'What a Compute-Centric Framework Says About Takeoff Speeds', *Open Philanthropy*, 27 June. Available at: <https://www.openphilanthropy.org/research/what-a-compute-centric-framework-says-about-takeoff-speeds/> (Accessed: 1 April 2025).

- 54 Narayanan, A. and Kapoor, S. (2025) *AI as Normal Technology*, Knight First Amendment Institute. Available at: <http://knightcolumbia.org/content/ai-as-normal-technology> (Accessed: 22 April 2025).
- 55 Bresnahan, T.F. and Trajtenberg, Ma. (1992) *General Purpose Technologies: 'Engines of Growth?'* 4148. Cambridge, MA: National Bureau of Economic Research. Available at: https://www.nber.org/system/files/working_papers/w4148/w4148.pdf (Accessed: 3 April 2025).
- 56 Helpman and Trajtenberg, 1996 (see n. 11)
- 57 Eloundou, T. et al. (2024) 'GPTs are GPTs: Labor market impact potential of LLMs', *Science*, 384(6702), pp. 1306–1308. Available at: <https://doi.org/10.1126/science.adi0998>.
- 58 Helpman and Trajtenberg, 1996 (see n. 11)
- 59 Brynjolfsson, 1993 (see n. 10)
- 60 Narayanan and Kapoor, 2025 (see n. 53)
- 61 Briggs, J. and Kodnani, D. (2024) *AI is showing 'very positive' signs of eventually boosting GDP and productivity*, Goldman Sachs. Available at: <https://www.goldmansachs.com/insights/articles/AI-is-showing-very-positive-signs-of-boosting-gdp> (Accessed: 3 April 2025).
- 62 Kokotajlo, D. et al. (2025) *AI 2027*. Available at: <https://ai-2027.com/> (Accessed: 3 April 2025).
- 63 Erdil, E. and Besiroglu, T. (2024) 'Explosive growth from AI automation: A review of the arguments'. arXiv. Available at: <https://doi.org/10.48550/arXiv.2309.11690>.
- 64 Gobble, M.M. (2018) 'Digitalization, Digitization, and Innovation', *Research-Technology Management*, 61(4), pp. 56–59. Available at: <https://doi.org/10.1080/08956308.2018.1471280>.
- 65 Gandhi, P., Khanna, S. and Ramaswamy, S. (2016) *Which Industries Are the Most Digital (and Why)?*, *Harvard Business Review*. Available at: <https://hbr.org/2016/04/a-chart-that-shows-which-industries-are-the-most-digital-and-why> (Accessed: 4 April 2025).
- 66 Manyika, J. et al. (2015) *Digital America: A tale of the haves and have-mores*, McKinsey Global Institute. Available at: <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/digital-america-a-tale-of-the-haves-and-have-mores> (Accessed: 4 April 2025).
- 67 Gandhi, Khanna and Ramaswamy, 2016 (see n. 64)
- 68 Olmstead, L. (2024) 'Digital Transformation & Tech Adoption by Sector', *The Whatfix Blog | Drive Digital Adoption*, 21 February. Available at: <https://whatfix.com/blog/digital-transformation-by-sector/> (Accessed: 4 April 2025).
- 69 Gandhi, Khanna and Ramaswamy, 2016 (see n. 64)
- 70 Sabbagh, K. et al. (2013) 'Digitization for Economic Growth and Job Creation: Regional and Industry Perspectives'. Available at: https://www3.weforum.org/docs/GITR/2013/GITR_Chapter1.2_2013.pdf.
- 71 Fulgenzi, R. (2025) 'An exploration of digitalization across multiple domains'. Available at: <https://iris.uniroma1.it/handle/11573/1733552> (Accessed: 4 April 2025).
- 72 Larson, J. et al. (2024) *The evolution of artificial intelligence (AI) spending by the U.S. government*, Brookings. Available at: <https://www.brookings.edu/articles/the-evolution-of-artificial-intelligence-ai-spending-by-the-u-s-government/> (Accessed: 4 April 2025).
- 73 Trachter, N. and Li, L. (2024) *Sectoral Multipliers and Technology Adoption as Insight Into Growth*, *Richmond Fed*. Available at: https://www.richmondfed.org/publications/research/economic_brief/2024/eb_24-27 (Accessed: 4 April 2025).
- 74 Dahlke, J. et al. (2024) 'Epidemic effects in the diffusion of emerging digital technologies: evidence from artificial intelligence adoption', *Research Policy*, 53(2), p. 104917. Available at: <https://doi.org/10.1016/j.respol.2023.104917>.
- 75 Cohen, W.M. and Levinthal, D.A. (1990) 'Absorptive Capacity: A New Perspective', in *Strategic Learning in a Knowledge Economy*. Hoboken: Taylor and Francis, p. 29.
- 76 Brock, J.K.-U. and von Wangenheim, F. (2019) 'Demystifying AI: What Digital Transformation Leaders Can Teach You about Realistic Artificial Intelligence', *California Management Review*, 61(4), pp. 110–134. Available at: <https://doi.org/10.1177/1536504219865226>.
- 77 Agrawal, A., Gans, J. and Goldfarb, A. (2019) 'Economic Policy for Artificial Intelligence', *Innovation Policy and the Economy*, 19, pp. 139–159. Available at: <https://doi.org/10.1086/699935>.
- 78 Goldfarb, A. and Tucker, C. (2012) 'Privacy and Innovation', *Innovation Policy and the Economy*, 12, pp. 65–90. Available at: <https://doi.org/10.1086/663156>.
- 79 Kinder, M. et al. (2024) *Generative AI, the American worker, and the future of work*, Brookings. Available at: <https://www.brookings.edu/articles/generative-ai-the-american-worker-and-the-future-of-work/> (Accessed: 2 April 2025).
- 80 Dahlke et al., 2024 (see n. 73)
- 81 McKinsey & Company, 2024 (see n. 32)
- 82 McElheran, K. et al. (2024) 'AI adoption in America: Who, what, and where', *Journal of Economics & Management Strategy*, 33(2), pp. 375–415. Available at: <https://doi.org/10.1111/jems.12576>.
- 83 McElheran et al., 2024 (see n. 82)
- 84 Muro, M., Methkuppally, S. and Kinder, M. (2025) *The geography of generative AI's workforce impacts will likely differ from those of previous technologies*, Brookings. Available at: <https://www.brookings.edu/articles/the-geography-of-generative-ais-workforce-impacts-will-likely-differ-from-those-of-previous-technologies/> (Accessed: 26 March 2025).

REFERENCES

- 85 Debevoise & Plimpton (2021) *Regulatory Risks for Not Disclosing Trading Algorithms - Five Takeaways from the SEC's \$170 million Settlement with BlueCrest Capital*, Debevoise. Available at: <https://www.debevoise.com/insights/publications/2021/01/regulatory-risks-for-not-disclosing-trading> (Accessed: 20 May 2025).
- 86 Garg, A. et al. (2024) *The state of retail banking: Profitability and growth in the era of digital and AI* | McKinsey, McKinsey & Company. Available at: <https://www.mckinsey.com/industries/financial-services/our-insights/the-state-of-retail-banking-profitability-and-growth-in-the-era-of-digital-and-ai> (Accessed: 20 May 2025).
- 87 McCoy, B., McCarthy, S. and McLaughlin, Z. (2020) *Regional Banking Technology and the Cloud*, Deloitte United States. Available at: <https://www2.deloitte.com/us/en/pages/consulting/articles/regional-banking-technology-and-the-cloud.html> (Accessed: 21 May 2025).